LENSI GONDALIYA

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RESEARCH INTERESTS

Data-driven modeling and machine learning for complex systems, including reliability analysis, failure prediction, and maintenance optimization, with applications in healthcare and engineering systems.

EDUCATION

Bridgewater State University — Bridgewater, MA
B.S. Computer Science, Minor in Management (GPA: 3.49/4.0) — Expected Dec. 2026

Bunker Hill Community College — Boston, MA · 2023 – 2024
University of Massachusetts Boston — Boston, MA · 2022 – 2023

AWARDS & HONORS

Audience Choice for Best Presentation, Student Arts & Research Symposium (STARS), BSU — 2026
Horace Mann Merit Scholarship ($5,000/year), Bridgewater State University — 2025 – 2026
Dean's List, Bridgewater State University — 2025 – 2026
Adrian Tinsley Program (ATP) Summer Research Grant ($5,000), Bridgewater State University — 2025
Experimental Research Award, NEMI 2025 Workshop, Northeastern University — 2025
Dean's List, Bunker Hill Community College — 2024

RESEARCH

Undergraduate Researcher | Advisor: Dr. Poonam Kumari — Bridgewater, MA
Bridgewater State University, Department of Computer Science — Jan. 2026 – May 2026

Project Title: Enhancing Content Discovery on Streaming Platforms: An Exploratory Analysis of IMDb (with Dawens Fortune)

- Analyzed an 85,855-record, 22-attribute IMDb dataset to identify the factors driving film popularity and similarity
- Built a data-preprocessing pipeline (missing-value imputation, deduplication, outlier filtering) to prepare data for modeling
- Applied and compared supervised and unsupervised models (Decision Tree, KNN, Linear Regression, K-Means), using the elbow method to select K=14 for the KNN classifier
- Identified vote count and review volume as the strongest predictors of popularity through correlation analysis
- Deployed the trained classifier as an interactive Flask web application

Adrian Tinsley Program (ATP) Summer Research Fellow | Advisor: Prof. Haleh Khojasteh — Bridgewater, MA
Bridgewater State University — Summer 2025

Project Title: Climate Strike (with Karima Olmstead and Menud Rahim)

- Completed a full-time, faculty-mentored summer research project funded by a competitive $5,000 ATP grant
- Collected climate strike data from countries around the world with a student research team and conducted a comparative, cross-country analysis in Python
- Presented findings at the NEMI 2025 Workshop, Egan Research Center, Northeastern University (August 2025), receiving the Experimental Research Award

PRESENTATIONS

Gondaliya, L. & Fortune, D. *Enhancing Content Discovery on Streaming Platforms: An Exploratory Analysis of IMDb.* Student Arts & Research Symposium (STARS), Bridgewater State University, Bridgewater, MA, Oral Presentation: April 2026. *Audience Choice for Best Presentation.*

Gondaliya, L., Olmstead, K., & Rahim, M. *Climate Strike.* NEMI 2025 Workshop, Egan Research Center, Northeastern University, Boston, MA, Presentation: August 2025. *Experimental Research Award.*

PROFESSIONAL EXPERIENCE

ZIEN Medical Technologies, LLC — Boston, MA
System Analyst Intern — Jun. 2025 – Sep. 2025

- Conducted FMEA on medical devices, scoring the severity, occurrence, and risk of potential failure modes to support device reliability
- Collaborated with Quality Engineering to review findings and reported results to supervisor for validation

Bridgewater State University — Bridgewater, MA
Information Technology Analyst (Intern) — Jun. 2026 – Aug. 2026

- Served as a bridge between the university's business goals and its technical systems as part of an IT project team

- Tested new systems and software updates and supported their rollout, training and assisting staff on the new tools
- Analyzed IT data and reports and automated repetitive tasks to reduce manual work

PROJECTS

HealthChain: Open-Source Healthcare AI Framework GitHub
COMP 490 Senior Design & Development (Prof. John Santore) Spring 2026

- Contributed in a 3-person team to HealthChain, an open-source Python framework that connects AI and machine learning systems with healthcare data and electronic health record (EHR) systems using the FHIR standard
- Resolved issue #117 by improving validation logic and error messages for invalid FHIR resource selection formats (e.g., ResourceType/ID)
- Improved input validation in the order-select and order-sign clinical decision support workflows, testing fixes across multiple invalid inputs with pytest

Voice-Enabled AI Meeting Room Agent GitHub
COMP 490 Senior Design & Development 2026

- Defined product requirements independently for a voice-driven meeting assistant that answers spoken questions in real time
- Integrated a speech-to-text API to transcribe voice input and respond to user queries
- Deployed to DigitalOcean with an automated GitHub Actions CI/CD pipeline

2000s Nostalgia Movie Recommender
COMP 470 Introduction to Artificial Intelligence (group project) 2026

- Built a movie assistant that recommends 2000s films using a local language model and retrieval-augmented generation (RAG)
- Implemented labs on logistic regression, KNN, RAG evaluation, conversation memory, and tool routing

ACTIVITIES AND LEADERSHIP

The Comment (BSU Student Newspaper) Bridgewater, MA
Creative Director Jan. 2025 – Present

- Own end-to-end layout and design production in Adobe InDesign for 10 biweekly issues per semester, coordinating a 9-person editorial team
- Built a production calendar with buffer deadlines, reprioritizing content with the Editor-in-Chief and 7 section editors when sections fell behind
- Maintained a zero-late-issue record as the primary point of accountability for visual consistency and quality control

Information Technology Division, Bridgewater State University Bridgewater, MA
IT Outcomes Writer Jun. 2026 – Aug. 2026

- Interviewed 4 stakeholders per article to produce 2 published feature pieces, translating technical initiatives into clear narratives

Think Tank Club, Bridgewater State University Bridgewater, MA
Lab Technician Sep. 2025 – May 2026

- Trained 20–25 students one-on-one on safe operation of 3D printers and robotics equipment, troubleshooting calibration issues in real time

Women in STEM, Vice President · **Computer Science Club**, Member · Bridgewater State University

RELEVANT COURSEWORK

Mathematics & Statistics: Calculus I, II & III (Multivariable), Linear Algebra, Discrete Mathematics, Statistical Methods

Computing & Data: Data Structures & Algorithms, Analysis of Algorithms, Data-Intensive Computing, Introduction to Artificial Intelligence, Data Mining (in progress), SQL Programming, Software Engineering

Science & Management: General Physics I & II, Human Resources Management (in progress)

TECHNICAL SKILLS

Programming: Python, R, MATLAB, SQL, Java, C, C++, JavaScript, Go, HTML

Data Analysis & Machine Learning: pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn, Hugging Face Transformers, LangChain, LLM APIs (OpenAI, Ollama), RAG, Jupyter Notebook, Google Colab

Optimization, Simulation & Analytics: Pyomo, Arena, Simio, Minitab, Excel (Solver, pivot tables), Tableau, Power BI

Databases: MySQL, SQLite, Microsoft SQL Server, Oracle, MongoDB, Firebase

Development & Cloud: Flask, Django, FastAPI, Node.js, React.js, Streamlit, REST APIs, Git, GitHub Actions (CI/CD), Docker, Linux, AWS, Google Cloud Platform, Microsoft Azure, DigitalOcean

Methods & Tools: FMEA (reliability and risk analysis), pytest, JUnit, Agile/Scrum, Jira, Figma, Adobe Creative Suite